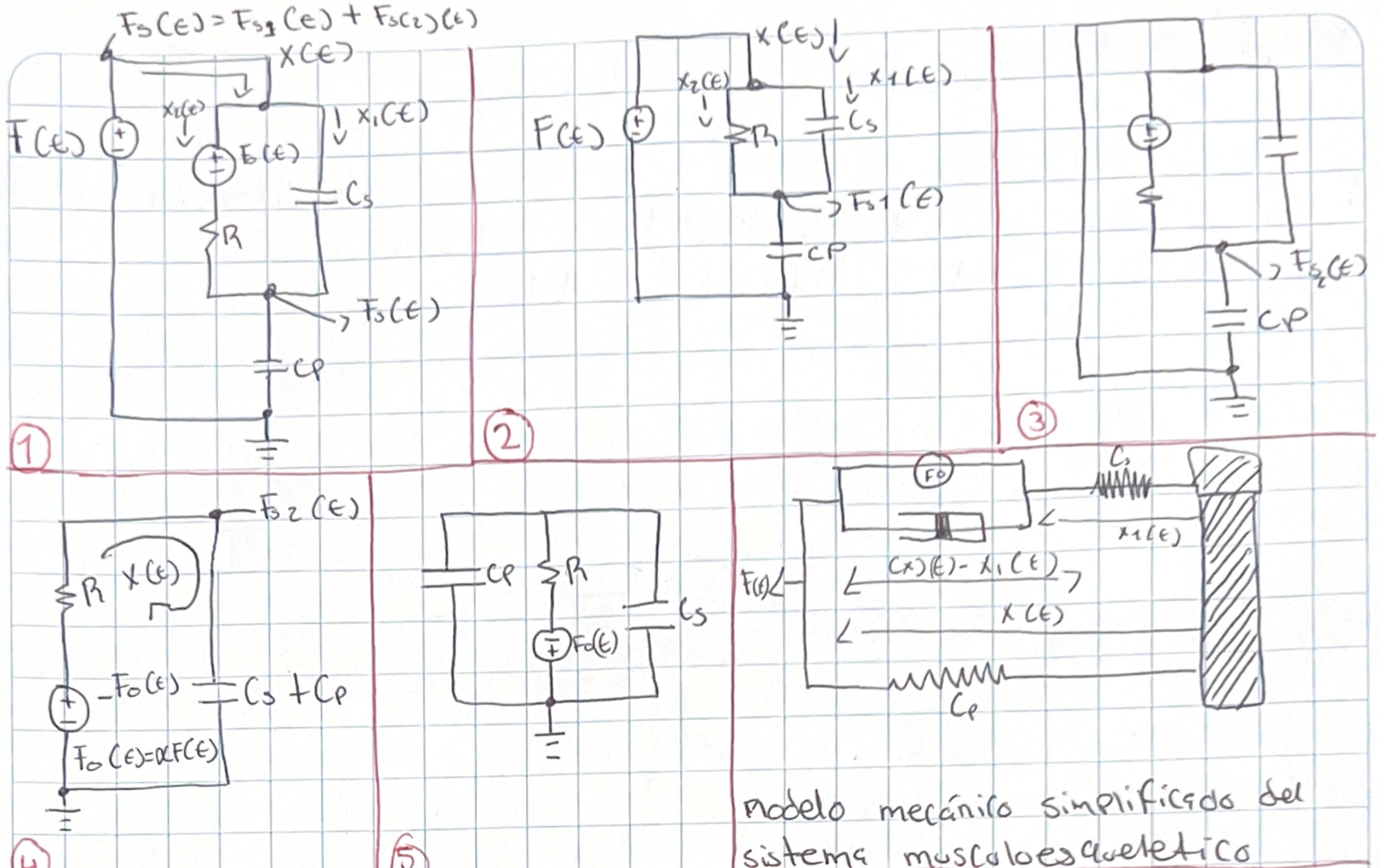
28/oct/26

Practica 3: Modelado y control del sistema musculoesqueletico



Modelo mecánico simplificado del sistema musculoesqueletico

Análisis del circuito por SuperPosición

Circuito #2: APagando $F_o(\epsilon)$

$$X(\epsilon) = x_1(\epsilon) + x_2(\epsilon)$$

$$x(\epsilon) = C_p \, dF_{s1}(\epsilon)/d\epsilon$$

$$x_1(\epsilon) = C_s \frac{d[F(\epsilon) - F_{s1}(\epsilon)]}{d\epsilon}$$

$$x_2(\epsilon) = \frac{F(\epsilon) - F_{s1}(\epsilon)}{R}$$

$$C_p \frac{dF_{s1}(\epsilon)}{d\epsilon} = C_s \frac{d[F(\epsilon) - F_{s1}(\epsilon)]}{d\epsilon} + \frac{F(\epsilon) - F_{s1}(\epsilon)}{R}$$ ← Primera ecuación

$$C_p S F_{s1}(s) = C_s S[F(s) - F_{s1}(s)] + \frac{F(s) - F_{s1}(s)}{R}$$

$$C_p S F_{s1}(s) = C_s S F(s) - C_s S F_{s1}(s) + \frac{F(s)}{R} - \frac{F_{s1}(s)}{R}$$ } Laplace

$$(C_p S + C_s S + \frac{1}{R}) F_{s1}(s) = (C_s S + \frac{1}{R}) \cdot F(s)$$ ← agrupar

$$\frac{R(C_p+C_s)s+1}{R}F_{s1}(s) = \frac{C_sR_s+1}{R}F(s)$$ * multiplicar R

$$F_{s1}(s) = \frac{\frac{C_sR_s+1}{R}F(s)}{\frac{R(C_p+C_s)s+1}{R}} = \frac{C_sR_s+1}{R(C_p+C_s)s+1}F(s)$$ * Primera salida

Circuito #4 Apagando F(t)

$$x(t) = C_p\frac{dF_{s2}(t)}{dt}$$

$$x(t) = C_s\frac{dF_{s2}(t)}{dt} + \frac{F_{s2}(t)-F_0(t)}{R}$$ * Primera ecuación

$$C_p = \frac{dF_{s2}(t)}{dt} = C_s\frac{dF_{s2}(t)}{dt} + \frac{F_{s2}(t)-F_0(t)}{R}$$

* LaPlace

$$C_pSF_{s2}(s) = -C_sSF_{s2}(s) + \frac{F_{s2}(t)-F_0(s)}{R}$$

$$C_pSF_{s2}(s) = -C_sSF_{s2}(s) + \frac{F_0(s)}{R} - \frac{F_{s2}(s)}{R}$$

$$\left(C_pS + C_sS + \frac{1}{R}\right)F_{s2}(s) = \frac{F_0(s)}{R}$$ * agrupar

$$[R(C_p+C_s)s+1]\,F_{s2}(s) = F_0(s)$$ * multiplicar R

$$F_{s2}(s) = \frac{1}{R(C_p+C_s)s+1}F_0(s)$$ * segunda salida?

Como lo hizo el Profe.

$$-\alpha F(t) = Rx(t) + \frac{1}{C_p+C_s}\int x(t)\,dt$$

$$F_{s2}(t) = \frac{1}{C_p+C_s}\int x(t)\,dt$$

$$-\alpha F(s) = RX(s) + \frac{X(s)}{C(C_p+C_s)s}$$

$$F_{s2}(s) = \frac{X(s)}{C(C_p+C_s)s}$$

$$F(s) = -\frac{X(s)}{\alpha}\cdot\frac{R(C_p+C_s)s+1}{C(C_p+C_s)s}$$

$$\frac{F_{S2}(S)}{F(S)} = \frac{\frac{X(S)}{(C_p+C_s)S}}{-\frac{X(S)}{\alpha}\,\frac{R(C_p+C_s)S+1}{(C_p+C_s)S}} = -\frac{\alpha}{R(C_p+C_s)S+1}$$

$$F_{S2}(S) = -\frac{\alpha F(S)}{R(C_p+C_s)S+1}$$

Función de transferencia

$$F_S(S) = F_{S1}(S) + F_{S2}(S) = \frac{C_s R_s + 1}{R(C_p+C_s)S+1}\,F(S) - \frac{\alpha F(S)}{R(C_p+C_s)S+1}$$